

EDUCATION

National Tsing Hua University

M.S. in Statistics and Data Science, GPA: 4.00/4.30

Hsinchu, Taiwan

2023–2025

- Thesis: “Nonparametric Mediation Analysis of Non-Markov Illness-Death Model”
- Advisor: Prof. Yen-Tsung Huang

National Chung Hsing University

B.S. in Applied Mathematics; Double Major in Finance, GPA: 4.10/4.30

Taichung, Taiwan

2018–2023

RESEARCH INTERESTS

Nonparametric methods; statistical learning theory; causal mediation analysis; survival analysis.

EXPERIENCE

Academia Sinica

Research Assistant

Taipei, Taiwan

2025–Present

- Supervisor: Prof. Yen-Tsung Huang
- Project: Multivariate Mediation Analysis via SIR-Based Single-Index Models

Developed a two-stage semiparametric framework for estimating causal mediation effects with multivariate mediators. The method combines sliced inverse regression for dimension reduction with two single-index models for estimation.

National Chung Hsing University

Teaching Assistant

Taichung, Taiwan

2021–2023

- Courses: Calculus and Ordinary Differential Equations

PUBLICATIONS

- [1] L.-S. Zhuang, J.-C. Yu, and Y.-T. Huang, “Non-Parametric Mediation Analysis of Non-Markov Illness-Death Model,” *Statistics in Medicine* 45, no. 18-19 (2026): e70685.

SCHOLARSHIPS AND AWARDS

- Poster Competition Winner, 2026 ISI–ISM–ISSAS Joint Conference, Academia Sinica 2026
- Prof. Chen Wen-Cheng Statistical Science Scholarship (陳文成統計科學獎學金) 2025
Received for outstanding master’s thesis research in statistical science; awarded annually to only two master’s students.
- Award of Excellence, The 7th Graduate Student Symposium on Statistical Research 2025
- Honorary Member of The Phi Tau Phi Scholastic Honor Society (中華民國斐陶斐榮譽學會榮譽會員) 2022
- Outstanding Teaching Assistant Award – Calculus 2021

CONFERENCE TALKS

- European Meeting of Statisticians, Lugano, Switzerland 2026
Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- South Taiwan Statistics Conference (南區統計研討會), Taipei, Taiwan 2025
Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- South Taiwan Statistics Conference, Kaohsiung, Taiwan 2024
Topic: Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process

POSTERS

- ISI-ISM-ISSAS Joint Conference, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2026
Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- The 7th Graduate Student Symposium on Statistical Research, Taichung, Taiwan 2025
Topic: Nonparametric Mediation Analysis of Non-Markov Illness-Death Model
- Poster Competition, Institute of Statistical Science, Academia Sinica, Taipei, Taiwan 2025
Topic: Causal Mediation of Semi-competing Risks Using Interventional Counterfactual Process

SELECTED PROJECTS

- A General Paired t -Test under Non-i.i.d. Assumptions 2024
Applied the Lyapunov Central Limit Theorem and k -nearest neighbors to extend the classical paired t -test framework.
- Fourier-Based Stock Price Modeling 2024
Applied the Fourier transform to denoise log returns in stock price data, improving prediction accuracy in a time series forecasting project.

TECHNICAL SKILLS

I have experience using R, Python, and Matlab for data analysis, model development, and statistical computing.